

POULAMI BURMAN

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EDUCATION

• Manipal Institute of Technology [MIT]	Udupi, India
• <i>B.Tech – Computer Science, Minor – Entrepreneurship</i>	2023 – 2027 (Expected)
• Loyola High School, Patna	CBSE
• <i>Class XII – 82%</i>	2023
• Carmel High School, Patna	ICSE
• <i>Class X – 93%</i>	2021

SKILLS SUMMARY

- **Machine Learning & NLP::** PyTorch, Transformers (Hugging Face), LLMs, NLP, Deep Learning
- **Languages::** C++, Python, Java, C
- **Core CS::** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks
- **Frameworks::** Node.js, Express.js, Flask, Django
- **Frontend::** HTML, CSS, Bootstrap, React
- **Databases::** MongoDB, MySQL, PostgreSQL
- **Tools::** Git, GitHub, Linux, REST APIs, Figma, Hugging Face, Kiro, Windsurf

PROJECTS

- **PurePick – AI-Powered Cosmetic Sustainability Scanner (Next.js, React, Tailwind CSS, Node.js, MongoDB):**
 - Built a full-stack web app enabling users to scan cosmetic barcodes and receive instant sustainability ratings, integrating Open Beauty Facts and EWG APIs to evaluate 50,000+ product entries in real-time.
 - Designed a serverless Next.js API layer on Vercel reducing average product analysis response time to under 1.2 seconds, covering ethics, ingredient safety, and eco-certification checks.
 - Followed Agile methodology with iterative feature releases, using Git for version control and collaborative development.
- **MediBot – Generative AI Medical Q&A Chatbot (Python, LangChain, Flask, Gemini API, Pinecone):**
 - Engineered a RAG-based chatbot using LangChain and Pinecone vector store, ingesting 500+ medical documents and enabling semantically accurate responses to patient health queries.
 - Integrated Gemini-2.5 with a custom prompt-safety layer to filter non-medical queries and reduce hallucination risk, achieving an 87% response relevance score across 200 test prompts.
 - Deployed as a Flask REST API with chunked document retrieval, reducing average query-to-response latency to under 2 seconds while maintaining context across multi-turn conversations.
- **Deep Learning-based Image Analysis System (PyTorch, OpenCV):**
 - Built a CNN-based image processing pipeline using PyTorch for face recognition and extraction
 - Integrated OpenCV preprocessing (filtering, edge detection) to improve input quality
 - Designed ROI-based processing to enhance efficiency and reduce computation
 - Evaluated and optimized model performance across test datasets

EXPERIENCE AND LEADERSHIP

- **President, ArtPod, MIT Manipal** Aug 2025 – Present
 - Directed a team of 100+ members, overseeing 10+ events with 150+ student participants; led digital outreach and brand campaigns to grow event visibility across platforms.
 - Established structured workflows for event planning and execution, improving team efficiency and coordination.
- **Marketing Head, Yaanbarpe** 2025 – Present
 - Leading marketing strategy for a growing startup, driving user acquisition and brand visibility.
 - Designed and executed digital campaigns, optimizing engagement and conversion metrics.
 - Collaborating with cross-functional teams to align product positioning with customer needs.
- **Executive, E-Cell MIT Manipal** Sept 2024 – May 2025
 - Coordinated the Manipal Entrepreneurship Summit 2025 (speakers: Dr. Velumani, Raj Shamani, Ashneer Grover), managing partnerships, sponsorships, logistics, and end-to-end marketing strategy spanning content, social media, and brand positioning.
 - Managed partnerships and sponsorships by crafting targeted outreach communication and pitch decks, coordinating with external stakeholders and internal teams to align on deliverables and timelines.

CERTIFICATIONS

- **Google AI Essentials – Coursera:** Completed Google’s foundational AI course covering core ML concepts, responsible AI practices, and practical applications of AI tools in the workplace.